

CMPS 4200 HCI Course: AI-Enhanced UI/UX

Project Requirements

Course Overview

This project-based course focuses on designing and developing user interfaces for applications that integrate artificial intelligence capabilities. Students will learn to create user-centered designs that effectively utilize pre-trained AI models from platforms like Hugging Face, OpenAI API, and other ML services to solve real-world problems.

Learning Objectives:

- Design intuitive user interfaces for AI-powered applications
 - Understand how to integrate pre-trained AI models effectively
 - Apply user-centered design principles to AI/ML systems
 - Consider ethical implications and user trust in AI applications
 - Develop skills in prototyping, user testing, and iterative design
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Phase 1: Project Concept, User Research & AI Model Exploration

Duration: 2 weeks

Weight: 15% of final grade

Objective

Students will form teams (2-3 members) and develop a comprehensive project concept that combines user needs with AI capabilities. This phase emphasizes understanding both your users and the AI tools available to serve them.

Key Deliverables

1. Project Proposal (35% of phase grade)

Requirements:

- **Problem Statement:** Clearly define the problem your application will solve
- **Target Users:** Identify your primary and secondary user groups
- **Core Functionality:** Outline 5-7 key features your application will provide
- **AI Integration Strategy:** Explain how AI will enhance the user experience (not just add complexity)
- **Success Metrics:** Define how you'll measure if your solution works

Example: A language learning app that uses speech recognition for pronunciation feedback and NLP for personalized lesson recommendations.

2. AI Model Research & Selection (35% of phase grade)

Requirements:

- **Model Exploration:** Research and evaluate at least 5 pre-trained models that could enhance your application
- **Technical Assessment:** For each model, document:
 - Input/output requirements
 - API limitations and rate limits
 - Response time expectations
 - Cost considerations (if applicable)
- **Final Model Selection:** Choose 1-2 models and justify your selection based on user needs and technical feasibility
- **Integration Plan:** Outline how these models will work within your user flow

Recommended Sources:

- Hugging Face Model Hub
- OpenAI API
- Google Cloud AI
- Azure Cognitive Services
- AWS AI Services

3. User Research & Personas (30% of phase grade)

Requirements:

- **User Interviews:** Conduct interviews with at least 6 potential users
- **Research Synthesis:** Identify key user needs, pain points, and goals
- **Persona Development:** Create 3 detailed user personas including:
 - Demographics and background
 - Goals and motivations
 - Pain points and frustrations
 - Technology comfort level
 - Relationship with AI/automation
 - Context of use scenarios

AI-Specific Considerations:

- How do users currently solve the problem your AI will address?
- What are their expectations and concerns about AI-powered solutions?
- What level of AI transparency do they need?